

Using AHP-Entropy method to explore the influencing factors of spatial demand of EVs public charging stations: A case study of Jinan, China

Qimeng Ren, Ming Sun^{*}

School of Landscape, Northeast Forestry University, Harbin, 150000, China

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ABSTRACT

In global sustainability efforts, electric vehicles (EVs) represent a burgeoning trend in the green transformation of the automotive sector. Widespread EVs adoption is closely associated with the availability of public charging stations (PCS). Understanding the factors that affect spatial demand for PCS is crucial for accurately measuring user needs and optimizing placement strategies. This study employs a geographical information system (GIS) to develop a spatial visualization evaluation index system including five dimensions: population distribution, traffic organization, infrastructure, land use, and regional economy. This research utilizes an AHP-Entropy method to rigorously analyze the weights of various spatial factors affecting the spatial demand of PCS for EVs. This findings indicate that: 1) Infrastructure (0.403) has the greatest impact on the spatial demand for PCS, followed by traffic organization (0.356), regional economy (0.129), land use (0.066), and population distribution (0.046). 2) Of the 14 indicators, the betweenness of road network (0.292) is the most important influencing factor. In addition, the scale of commercial centers (0.144), the scale of park views (0.104), nighttime light distribution (0.094), and the scale of transportation stations (0.090) are also key factors. By integrating both subjective and objective factors in spatial geography, this study offers a comprehensive evaluation of factors influencing spatial demand for PCS. The article concludes by proposing PCS spatial layout strategies from five dimensions, thereby promoting the further development of the EVs industry from a low-carbon perspective.

1. Introduction

In response to global concerns about the protection of the environment paired with the need for sustainable development, EVs have commenced replacing traditional fuel vehicles, becoming the key trend in green transportation. Traditional automotive fuels, obtained from petroleum refining, are slowly being replaced by low-carbon and renewable fuels (Adiya and Atiku, 2024; Abdellatief et al., 2023). Bio-fuels, produced by converting biomass into liquid fuels through processes such as pyrolysis, gasification, or fermentation, represent a significant alternative energy source (Mustafa et al., 2023). The technological advancements have also improved the efficiency of biofuels, while multiple nations' governments are interested in cultivating EVs development to address clean energy demands. EVs depend primarily on electric power, with this shift supported by the ongoing development of battery technology and increased shares of solar, wind, and other renewable energy sources (Burov et al., 2023). Therefore, automotive fuels are projected to transition from crude oil fuels to a diverse energy system that will incorporate biofuels and electric energy, and will be

more energy efficient and have a lower environmental impact (Rekhletskaia et al., 2022; Abdellatief et al., 2022). The proliferation of EVs heavily relies on establishing PCS. In China, as the EVs market expands, the demand for PCS is increasing in lockstep. Affected by the gradual replacement of fuel vehicles by EVs, China's urban layout has experienced comprehensive transformation wherein EVs have significantly changed the process of decarbonization in Chinese cities. Spatial demand for PCS refers to the distribution of people's demand regarding the utilization of PCS in urban space. Nevertheless, cities such as Jinan, identified as low-carbon and climate-resilient, face challenges in developing their EVs industry, including insufficient and unevenly distributed PCS. This impedes the extensive adoption of EVs and user willingness to purchase them. Therefore, an in-depth analysis of the factors influencing the spatial demand for PCS is crucial for optimizing their layout and exploring a new avenue for low-carbon urban development (Xu et al., 2023).

The spatial demand factors of PCS for EVs are interconnected with several domains of research. It has been extensively researched by scholars at home and abroad. Studies utilized mathematical algorithms

^{*} Corresponding author. School of Landscape, Northeast Forestry University, China.

E-mail address: sunming@nefu.edu.cn (M. Sun).

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including the point demand model (Chen et al., 2023; Namdeo et al., 2014; Hu et al., 2023; Tao et al., 2021), cut-off siting model (Xu et al., 2024), minimum queuing time model (Chen et al., 2020), and minimum driving distance model (Pan et al., 2021) to carry out siting research with fixed capacity. Following this, academics introduced sociological behavioral decision-making theory to estimate users' travel and charging patterns and predict the distribution of electric loads (Kavianipour et al., 2021; Yang et al., 2021). Nevertheless, with advancements in big data technology, the combination of charging data, caller location data, and vehicle movement data enhanced the accuracy of electricity consumption predictions (Hu et al., 2022; Yang et al., 2023; Cui et al., 2023). It is essential to ensure that the charging needs of users are satisfied by analyzing how users demonstrate patterns of travel and initiate charging. Various factors that affect the choices of the users on PCS include the time of charging, the cost, the location of the facility, the type of facility, the route chosen and the traffic of the route, and the duration of parking (He et al., 2022; Wang and Wang, 2023; Jiang et al., 2024; Zhang et al., 2022; Saeed et al., 2020). The layout planning that balances supply and demand is instrumental in ensuring that the provision of PCS's does not exceed supply-demand imbalances. Several researchers have constructed evaluation index systems whose basics are in the concept of the "demand-determined supply" model for determining the need to construct the PCS (Bian et al., 2017).

In addition, the following aspects are also important factors affecting the layout of PCS. PCS planning is a crucial aspect of the urban infrastructure, aiming at the identification of the patterns of usage by the residents as well as the service radius of living circles (Wang et al., 2017; Liu et al., 2020). With the city comprising various functional sites, the mode of charging, time, and location of charging have a significant effect (Xiong et al., 2021; Ge and Li, 2019; Wang and Xia, 2022). The network configuration in relation to the urban road infrastructure significantly affects the arrangement of the EVs PCS. With a proper understanding of the road networks and the routing of vehicles by analyzing road network accessibility and vehicle routing, researchers can identify the spatial charging pattern of EVs users (Gagarin and Corcoran, 2018; Yang et al., 2019; Zhou et al., 2024). A robust transport network not only enhances the accessibility of PCS and further investment in charging strategies, cultivating a virtuous cycle. Policies and measures taken by the government can cultivate the charging infrastructure development along with the development of PCS through financial incentives, tax reduction, legal rules, and technical standards. The level of consistency and continuity in policies is one of the defining factors that determine user acceptance and investment confidence (Zhu et al., 2019). The construction of a charging network necessitates the collaboration of the public and private sectors, and the involvement of the private sector necessitates specific supportive investments and partnerships. Efficient use of capital further enhances the coverage of services and quality of the PCS, thus leading to customer satisfaction (Li et al., 2019). The economic level directly influences the demand for the purchase of EVs and investments in PCS. More regions with faster economic growth primarily have a higher density of EVs and PCS. With economic advancement, consumer acceptance of EVs increases, thus resulting in the demand for its market services and infrastructure increasing (Jia et al., 2016).

In summary, previous studies have limitations. They often focus on single methods or factors like using math models or domain-specific analysis (Wang et al., 2022; Zhang et al., 2023; Deb et al., 2021), mainly on current situation and incremental adjustment of public charging facilities, lacking comprehensive multi-factor analysis from urban planning perspective to predict distribution areas. Although recent studies in the fields of geography, architecture and urban planning have considered spatial distribution and influencing factors of charging facilities with GIS providing a new view (Wang et al., 2023; Liu et al., 2024), problems remain. It is found that population (Carra et al., 2022), road network (He et al., 2020a), infrastructure (Li et al., 2023), user travel behavior (Ma et al., 2007), economy and employment (Gupta

et al., 2023), land use and other factors have an impact on the layout of charging facilities (Csizsár et al., 2019). This study creates a multi-dimensional evaluation index system, integrating multi-source data and spatial geography for quantitative and visual analysis of spatial requirements, which was seldom done before. In weight determination, the single subjective and objective method was used in the past. This research innovatively introduces AHP-entropy method combined subjective and objective aspects, avoiding the problem that subjective arbitrariness and objective method ignore user needs. It could assign complex index weights more accurately. Compared with the previous single method, it could provide more accurate weights for better layout planning and promote research development.

Taking the urban area of Jinan, China as an example, this study applied AHP-entropy method to build a comprehensive evaluation index system from five dimensions: population distribution, traffic organization, infrastructure, land use and regional economy. In order to provide a basis for optimizing the layout of PCS, the research explored relevant influence mechanism and weight relationship of each factor on PCS spatial demand. This study aims to achieve the following goals: First, build a comprehensive evaluation index system to accurately identify the core factors that affect PCS spatial demand; Secondly, optimize the layout of PCS to improve its utilization rate and service quality to meet users' charging needs. Third, promote the development of EVs industry and provide new ideas for urban transportation to low-carbon sustainable transformation. The study framework is demonstrated in Fig. 1.

2. Study area and methodology

2.1. Study area

This paper evaluates Jinan, China, with a specific focus on its main urban area, bounded by the Jiguang Expressway, the Jingtai Expressway, and the Jinan Ring Expressway. This urban area incorporates a multitude of functions, including commercial businesses, transportation, cultural tourism and leisure opportunities, and essential services. It acts as the nucleus of urban vitality in Jinan City, this area attracts complex and diverse flows of both people and vehicles. A convergence of factors has led to this area becoming a central hub for the utilization of EVs and PCS. In contrast, areas outside the core, such as southern mountainous regions, water bodies, and rural areas, have low population density, poor road infrastructure, and incomplete support systems. Thus, their spatial vitality is low, and user demand for EVs and PCS is weak. So, these areas are excluded from this study (Sun and Ren, 2022). Since Jinan's EV industry and PCS infrastructure are still developing, analyzing factors influencing spatial demand evaluation is crucial for optimization, as shown in Fig. 2.

2.2. Research methods

In assessing the factors contributing to the spatial demand for EVs PCS, a thorough analysis integrating resident demand preferences with the characteristics of evaluative indicators is essential. Common approaches to evaluating such influential factors are in two categories: subjective and objective weighting methodologies (Kaya et al., 2020).

Subjective weighting methods, such as the Delphi method and AHP, determine the relative importance of each indicator based on the expert opinions and practical insights of specialists, researchers, and policy makers. When analyzing the spatial demand for EVs PCS, AHP offers a viable approach to weighting influential factors. This method allows for the comprehensive integration of user-specific requirements and broader public priorities. However, the potential for individual biases to influence outcomes introduces a degree of subjective variability into the assessment. Therefore, combining subjective methods with objective weighting strategies promotes more robust evaluative results.

Objective assignment methods, including principal component analysis, the entropy method, and the entropy weight method, leverage

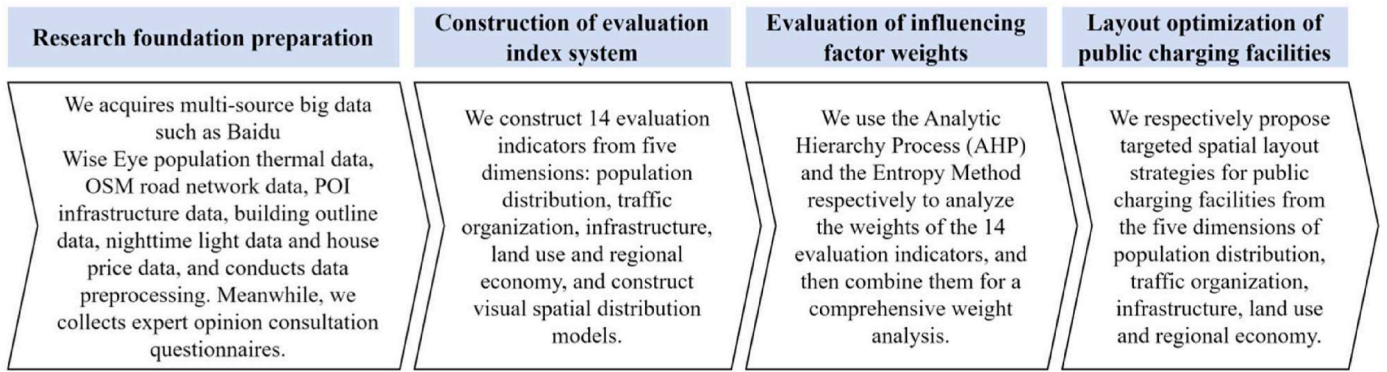


Fig. 1. Study framework.

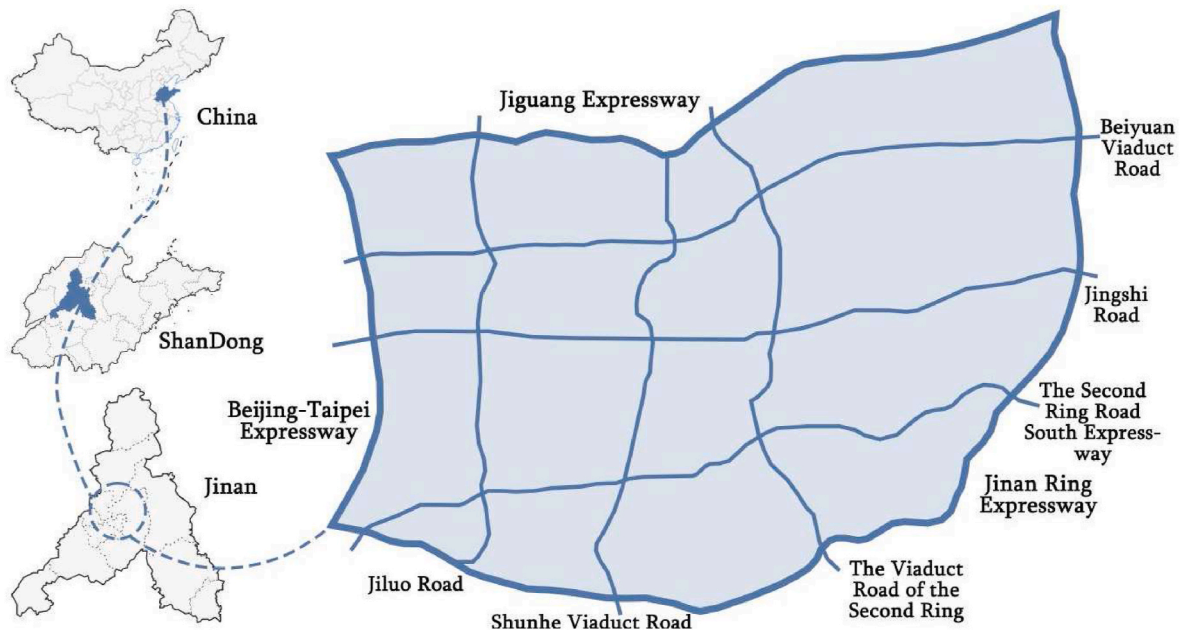


Fig. 2. Research location and scope definition.

objectively existing data and empirical information to determine the influence weights of individual indicators, thus promoting objectivity, reasonableness, and accuracy in the analysis results. This study also employs the entropy method to determine the weights of factors influencing the spatial demand for PCS for EVs. This approach relies entirely on objective data and seeks to offset the potential subjectivity in AHP. However, entropy-derived weights may not fully reflect user group preferences and may overlook subjective assessments of indicator importance relative to user needs (Zhao et al., 2023).

Therefore, this study utilizes the AHP-Entropy method. First, AHP and the entropy method are applied independently to the same evaluation system. Then, the results from these two analyses are integrated to establish the final composite weight (Zhang et al., 2022). This approach synthesizes the subjective nature of AHP and the objective nature of the entropy method. It allows for the representation of actual user demand preferences while simultaneously capitalizing on the information contained in the data itself, thus generating a holistic evaluation of influencing factors.

2.2.1. Analytic hierarchy process (AHP)

AHP facilitates decision-making in complex, multi-tiered indicator systems utilizing a hierarchical framework. This process breaks down the overarching objective into multiple criteria or indicator levels. By

applying fuzzy measurement to qualitative indicators, AHP determines the relationships between these levels. Then, the method calculates the contribution (weight) of each element in a given level to the elements in the level above. A weighted summation then derives the overall weight relative to the primary goal (Xie et al., 2019).

Firstly, a hierarchical structural model of the rating indicators is developed, including a target level (U), criteria levels (U_i), and indicator levels (U_{ij}). Following this, a pairwise comparison of rating indicators in each level is performed, employing a consistent matrix approach to construct the judgment matrix A. This matrix is represented as follows:

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \dots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix} \quad (1)$$

Where a_{ij} represents the importance assigned to the comparison between indicator i and indicator j ($i, j = 1, 2, 3, \dots, n$). Typically, multiple experts are consulted to individually assess the elements of each level's evaluation indicators. The relative importance of each indicator is established utilizing a 9-point scale, determined by values from 1 to 9 and their reciprocals, as detailed in Table 1 (Feng and Wang, 2023).

Following this, the geometric mean of the n elements in each row of

Table 1
Meaning of AHP scaling method.

Scale	Meaning
1	They have the same importance.
3	The vertical indicator (Fi) is slightly more important than the horizontal indicator (Fj).
5	The vertical indicator (Fi) is significantly more important than the horizontal indicator (Fj).
7	The vertical indicator (Fi) is considerably more important than the horizontal indicator (Fj).
9	The vertical indicator (Fi) is extremely important compared to the horizontal indicator (Fj).
Middle value (2,4,6,8)	The intermediate value between the above adjacent judgment scales
The reciprocal of 1–9	If the comparison score of Fi and Fj is Fij, then the comparison score of Fj and Fi is Fji

the judgment matrix is computed utilizing the following formula:

$$W_i = \frac{1}{n} \sum_{j=1}^n \frac{a_{ij}}{\sum_{k=1}^n a_{kj}} \quad (2)$$

Then, data normalization is performed to derive the relative weight vector for each evaluation index. The calculation is executed utilizing this formula:

$$W_i = \bar{W} / \sum_{i=1}^n \bar{W}_i \quad (3)$$

Next, a consistency check is conducted. First, the maximum eigenvalue $\lambda_{\max}$ of the judgment matrix needs to be calculated utilizing the following formula:

$$\lambda_{\max} = \frac{1}{n} \sum_{i=1}^n \frac{(BW)_i}{W_i} \quad (4)$$

The consistency index (C.I.) and the consistency ratio (C.R.) are then determined. To measure the C.I. at various levels, the random consistency index (R.I.) is employed, as presented in Table 2. Generally, C.R. has an inverse relationship with the consistency of the discriminant matrix. A C.R. value below 0.1 signifies that the data satisfies the consistency test; whereas, if $C.R. \geq 0.1$, the matrix requires re-evaluation (Wu et al., 2023). These calculations are performed as follows:

$$C.I. = \frac{\lambda_{\max} - n}{n - 1} \quad (5)$$

$$C.R. = \frac{C.I.}{R.I.} \quad (6)$$

Finally, the comprehensive weight W_i is calculated, synthesized hierarchically through the relative weights of each layer of evaluation indicators. This comprehensive weight represents the weight relative to the overall target layer. The calculation is as follows:

$$W = WU_i \times WU_{ij} \quad (7)$$

2.2.2. Entropy method

The entropy method determines weight based solely on the degree of dispersion exhibited by the numerical data values. The weight assigned to an indicator is inversely proportional to its information entropy and directly proportional to its discreteness, the information range it provides, and its influence in the comprehensive evaluation. The weight derived by the entropy method does not reflect the importance of the indicator's meaning, but rather the relative intensity of each indicator,

in a competitive sense, when the values of the evaluation indicators are fixed (Gu et al., 2020). According to Shannon's information entropy theory, the information entropy of a dataset is defined as $E_j = \ln(n)^{-1} \sum_{i=1}^n x_{ij} \ln x_{ij}$. The formula for calculating the corresponding weight is:

$$D_i = \frac{1 - E_j}{k - \sum E_j} \quad (i = 1, 2, \dots, k) \quad (8)$$

In the formula, D_i represents the calculation result of the index weight; E_j denotes the information entropy of the index data; x_{ij} expresses the standardized value of the original data. If it is a positive indicator, then $x_{ij} = \frac{e_{ij} - \min e_{ij}}{\max e_{ij} - \min e_{ij}}$, if it is a negative indicator, then $x_{ij} = \frac{\max e_{ij} - e_{ij}}{\max e_{ij} - \min e_{ij}}$; among them, e_{ij} , $\min e_{ij}$, $\max e_{ij}$ refer to the original index data, the minimum value of the index, and the maximum value of the index respectively (Ren and Sun, 2023).

2.2.3. AHP-entropy method

After calculating the weights for the evaluation index system of the spatial demand for PCS of EVs utilizing both the AHP and entropy methods, the respective calculation results must be combined to determine the comprehensive weight of the index's influencing factors (Yu et al., 2025; He and Zhao, 2024). The calculation formula is as follows:

$$W_j = \frac{\sqrt{\alpha_j \beta_j}}{\sum_{j=1}^n \sqrt{\alpha_j \beta_j}} \quad (9)$$

In the formula, W_j represents the comprehensive weight of the index; α_j represents the subjective weight obtained by AHP; β_j represents the objective weight obtained by entropy method.

2.3. Build an evaluation index system

A comprehensive review of multi-domain studies concerning the spatial layout of PCS, including both domestic and international research, along with practical experience and analysis of current conditions, indicates ten factors influencing the spatial demand for PCS of EVs: population, transportation, infrastructure, economy, land use, service area, construction plans, electricity, policy, and user demand. Specifically, population distribution, traffic organization, infrastructure, land use, and regional economic can be integrated with spatial geography through the collection and analysis of relevant data. This integration allows for the development of an evaluation index system designed to assess the spatial demand for EVs PCS. The analysis of existing PCS service areas, utilizing current PCS data, forms a foundational element of the evaluation, further informed by user demand data acquired through questionnaire surveys. Due to present data limitations, construction plans, power grid conditions, and policy frameworks cannot be presently integrated into the spatial geography analysis; therefore, these factors are excluded from consideration in the present evaluation index system, as depicted in Fig. 3.

In conclusion, the evaluation index system for the spatial requirements of PCS of EVs is organized hierarchically across three levels. The first level, designated as the target level (U), establishes the evaluation of spatial demand for PCS of EVs as its primary objective. The second level, the criterion level (U_i), is categorized into five dimensions: (U1) population distribution, (U2) traffic organization, (U3) infrastructure, (U4) land use, and (U5) regional economy. The third level, the index level (U_{ij}), further divided this framework into 14 specific evaluation sub-indicators, as presented in Table 3.

3. Results and analysis

3.1. The construction of spatial demand evaluation indicators

The construction of PCS spatial demand evaluation index is not only

Table 2
Average random consistency index R.I.

n	1	2	3	4	5	6	7	8
R.I	0	0	0.58	0.90	1.12	1.24	1.32	1.41

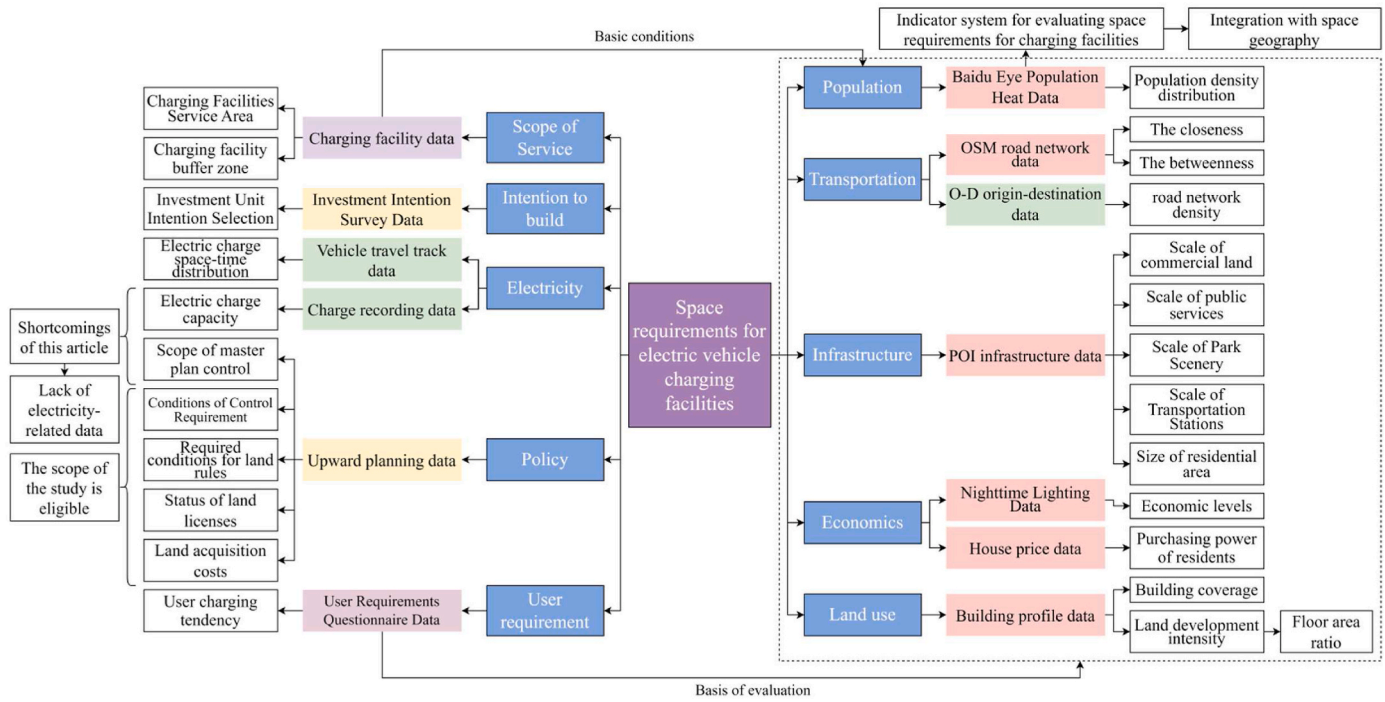


Fig. 3. The source of the evaluation indicators selected and the related data involved.

Table 3

Spatial demand evaluation index system of EVs PCS and its data and analysis method.

(U)Target level	(Ui)Criterion level	(Uij)Index level	Methodology for quantifying indicators	Data	Data sources
(U)Evaluation of spatial demand for PCS of EVs	(U1) Population distribution	(U11)Population vitality intensity on weekday	Average population thermal value on weekdays	Baidu Wise Eye population heatmap data	Baidu Wise Eye platform
		(U12)Population vitality intensity on rest day	Average population thermal value on rest days		
	(U2) Traffic organization	(U21)The closeness of road network	The closeness of SDNA	OSM road network data	OSM open-source map
		(U22)The betweenness of road network	The betweenness of SDNA		
	(U3) Infrastructure	(U23)Density of road network	Line density of road network	POI infrastructure data	AMAP Open Platform
		(U31)The scale of commercial centers	Density distribution of commercial facilities		
		(U32)The scale of public service facilities	Density distribution of public service facilities		
		(U33)The scale of park views	Density distribution of green spatial facilities		
		(U34)The scale of transportation stations	Density distribution of transportation facilities		
	(U4) Land use	(U35)The scale of residential neighborhoods	Density distribution of residential facilities	Building outline data	Atlas of heaven
		(U41)Building coverage	Building land area/total land area		
	(U5) Regional economy	(U42)Floor area ratio	Gross floor area/gross land area	Nighttime Lighting Data	NPP-VIIRS remote sensing imagery
		(U51)Nighttime light distribution	ArcGIS Spatial Connectivity		
		(U52)House price level	ArcGIS Spatial Connectivity	House price data	Real estate company

qualitative description, but also visualization in geographical space by GIS spatial analysis method. The quantitative analysis data of each index in the urban space of Jinan city can be obtained. In the subsequent weight analysis, the visual analysis results can be combined for subjective evaluation, and the quantitative data can be combined for objective evaluation. This study employed Baidu Wise Eye population heatmap data, collected over a 24-h period on both a weekday (Wednesday, September 27, 2023) and a weekend day (Saturday, September 23, 2023) in the urban area of Jinan City. This data was subjected to kernel density analysis in a GIS environment. Following aggregation to the

block spatial unit level and calculation of the average value, weekday population activity intensity (U11) and weekend population activity intensity (U12) were obtained. Next, after conducting the closeness and betweenness of SDNA analysis on the road network in the urban area of Jinan City and connecting to the block spatial unit, as well as calculating the result of the road network length/total block area within the block spatial, the closeness of road network (U21), the betweenness of road network (U22), and density of road network (U23) were obtained. The results of these five evaluation indicators, obtained from geospatial analysis, are displayed in Fig. 4.

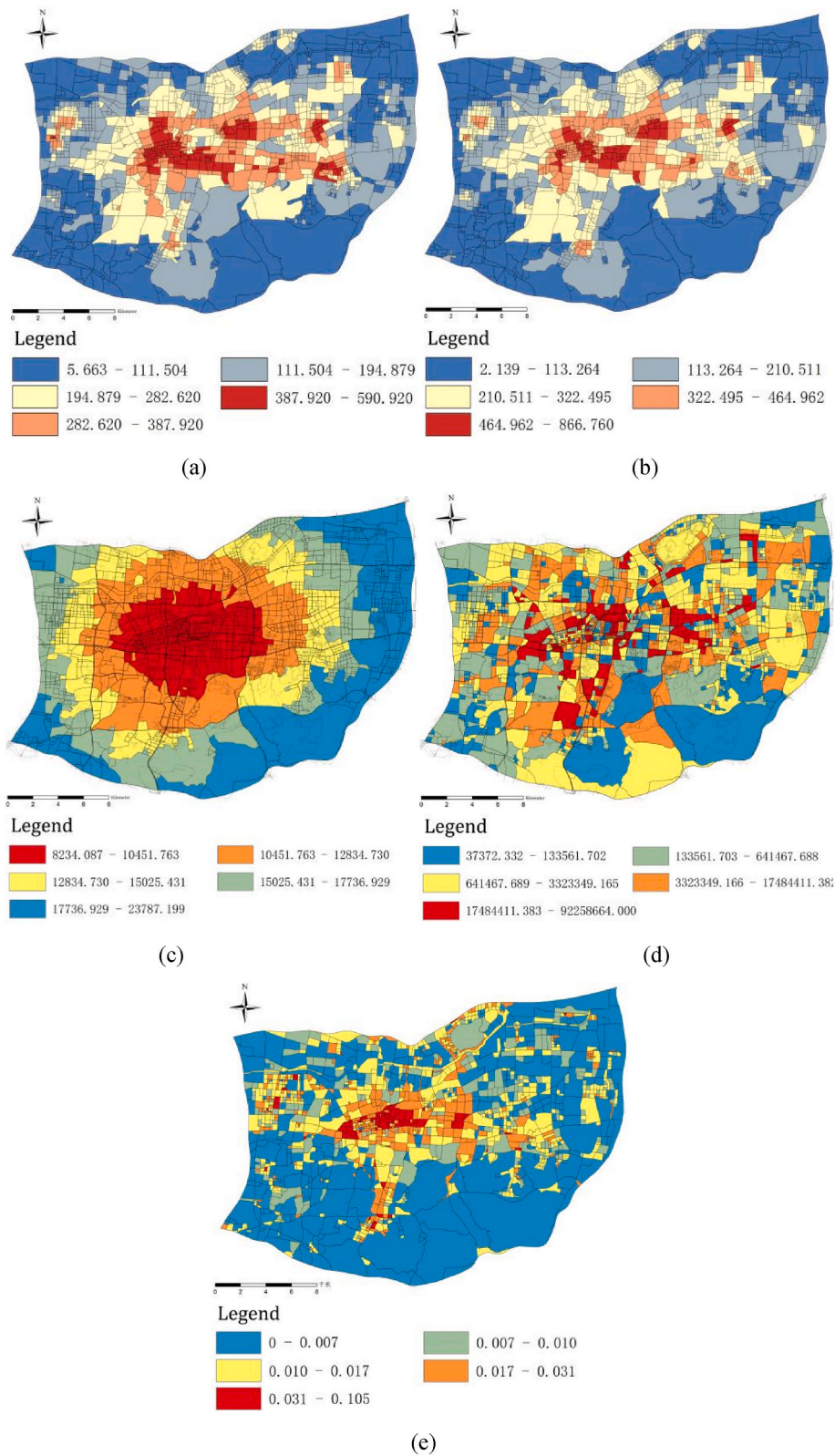


Fig. 4. Evaluation indicators of population distribution dimension and traffic organization dimension.

Note: (a) Population vitality intensity on weekday (U11); (b) Population vitality intensity on rest day (U12); (c) The closeness of road network (U21); (d) The betweenness of road network (U22); (e) Density of road network (U23).

This paper also conducts kernel density analysis of the five types of reclassified infrastructure data in GIS, namely commercial service land, land for public management and public service facilities, land for roads and transportation facilities, residential land, and land for green spaces and squares. Then, they are spatially connected to the block spatial units to obtain the scale of commercial centers (U31), the scale of public service facilities (U32), the scale of park views (U33), the scale of transportation stations (U34), and the scale of residential neighborhoods (U35). Moreover, by calculating the results of the construction land area/total land area and the total building area/total land area in the block space, the building coverage (U41) and the floor area ratio (U42) are obtained. Finally, the nighttime light data and housing price data obtained are spatially connected to the block spatial units through GIS to obtain nighttime light distribution (U51) and housing price level (U52). Fig. 5 demonstrates the results of the nine evaluation indicators obtained after geospatial analysis of the data.

3.2. Analysis of influencing mechanism of evaluation indicators

In order to confirm the research question and hypothesis of this paper, the spatial demand evaluation index system of PCS is correlated with the spatial layout of PCS. Then AHP-entropy method can be used

for further exploration, and finally the layout of PCS can be determined by spatial demand analysis model. In this paper, the "bivariate correlation analysis" method in SPSS was used to conduct Spearman correlation analysis on the spatial demand evaluation indicators of 14 EVs PCS and the distribution of current PCS. The significance test was a two-tail test, are displayed in Table 4. The analysis results show that there is a significant positive correlation between the 14 spatial demand evaluation indicators and the current layout of PCS, with the significance reaching 0.000. By analyzing the correlation coefficients, it can be found that the correlation coefficients of (U11) Population vitality intensity on week-day, (U12) Population vitality intensity on rest day, (U34) The scale of transportation stations and (U35) The scale of residential neighborhoods are all greater than 0.7, showing a strong correlation; The correlation coefficients of (U21) The closeness of road network, (U23) Density of road network, (U31) The scale of commercial centers, (U32) The scale of public service facilities, (U33) The scale of park views and (U51) Nighttime light distribution were between 0.5 and 0.7, showing a strong correlation. However, the correlations of (U22) The betweenness of road network, (U41) Building coverage, (U42) Floor area ratio and (U52) House price level were all lower than 0.5, indicating a relatively weak correlation. It can be seen that the spatial layout of PCS is more affected by population activities, transportation, and daily residential life. This

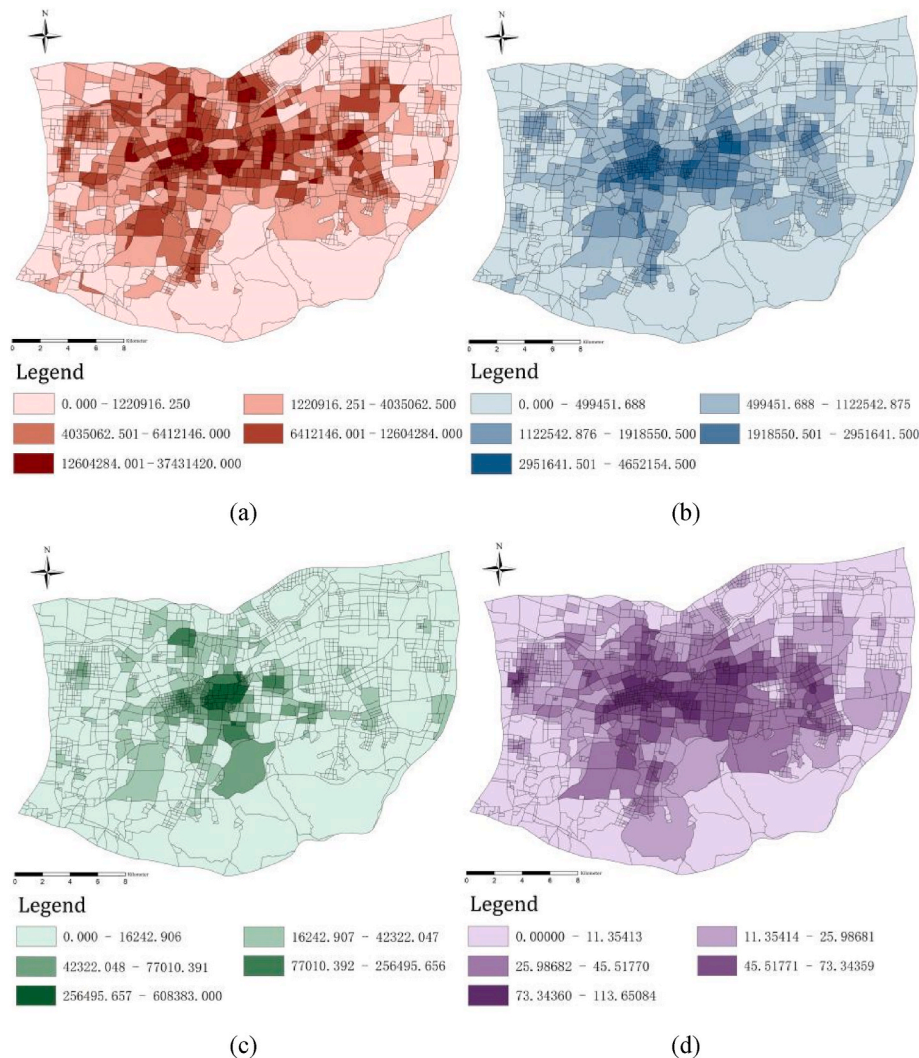


Fig. 5. Evaluation indicators of infrastructure dimension, land use dimension and regional economic dimension.

Note: (a) The scale of commercial center (U31); (b) The scale of public service facility (U32); (c) The scale of park view (U33); (d) The scale of transportation station (U34); (e) The scale of residential neighborhood (U35); (f) Building coverage (U41); (g) Floor area ratio (U42); (h) Nighttime light distribution (U51); (I) Housing price level (U52).

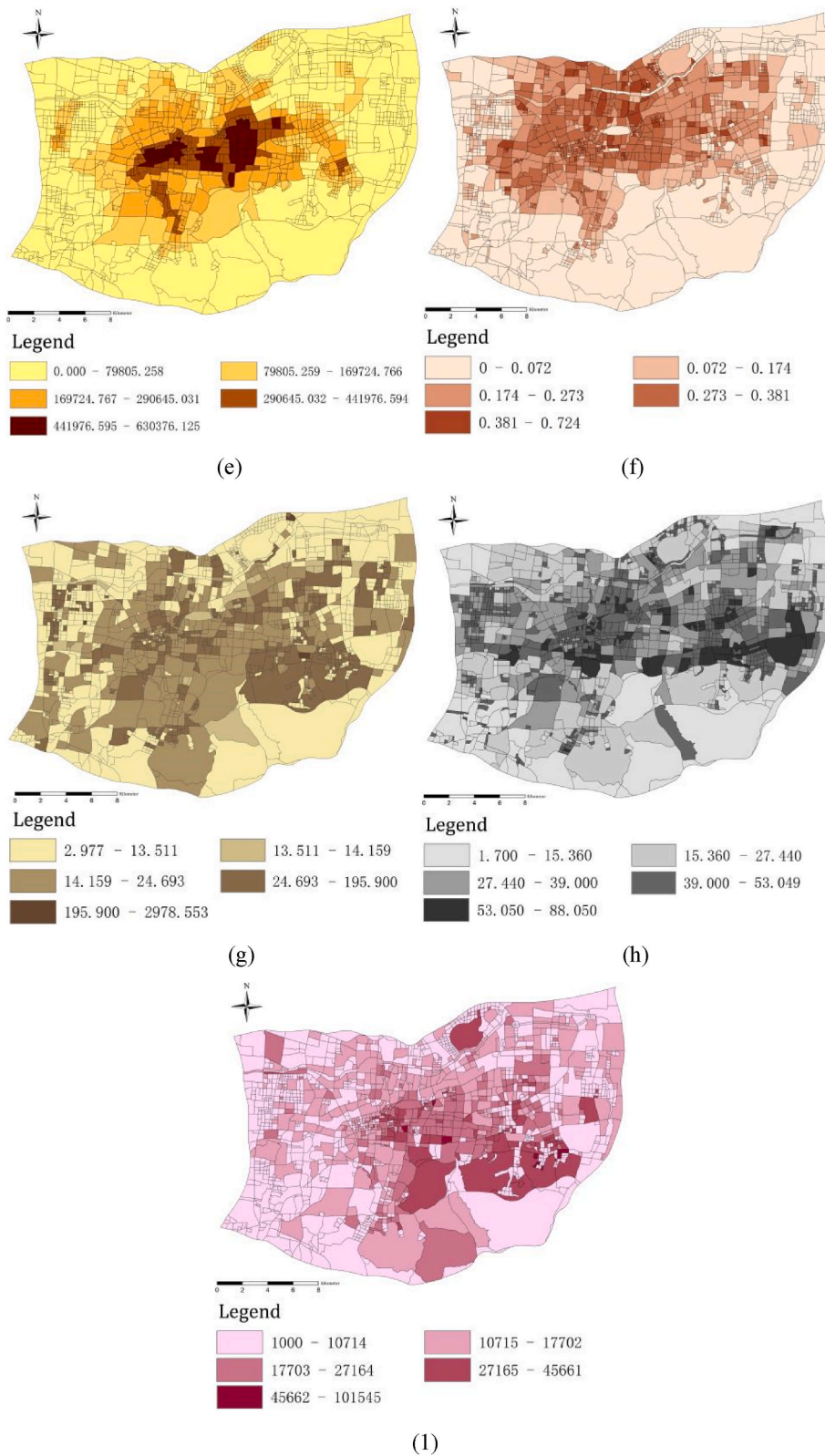


Fig. 5. (continued).

means that PCS tend to be located in areas with high population activity. And this layout is closely related to residents' travel patterns. On the contrary, the traffic flow, building layout and housing price level are weakly correlated with the layout of PCS.

3.3. Determination of the weights of evaluation indicators of AHP

3.3.1. Construct judgment matrix to calculate relative weight and consistency test

In this study, the relative importance of space demand evaluation index of PCS was discussed by means of questionnaire scoring. The

Table 4

Correlation analysis between spatial demand evaluation index and current PCS layout.

Spatial demand evaluation indicators	Current layout of PCS	Spatial demand evaluation indicators	Current layout of PCS
(U11)Population vitality intensity on weekday	0.742**	(U33)The scale of park views	0.502**
(U12)Population vitality intensity on rest day	0.727**	(U34)The scale of transportation stations	0.762**
(U21)The closeness of road network	0.692**	(U35)The scale of residential neighborhoods	0.709**
(U22)The betweenness of road network	0.253**	(U41)Building coverage	0.142**
(U23)Density of road network	0.597**	(U42)Floor area ratio	0.290**
(U31)The scale of commercial centers	0.636**	(U51)Nighttime light distribution	0.652**
(U32)The scale of public service facilities	0.602**	(U52)House price level	0.386**

Note: **. The correlation was significant at layer 0.01 (double tail).

source of the questions is the evaluation index system established above, and the chapters of the questions are presented in the judgment matrix scoring table constructed by pair-to-pair comparison of the indicators of the criterion layer and the indicator layer respectively (refer to Table 4, Table 5, Table 6, Table 7, Tables 8 and 9). Thirty questionnaires were distributed to experts, scholars, researchers, and urban planners specializing in urban planning, as well as to EVs owners with more than five years of driving experience. Following the AHP scoring standards and structural model, these evaluators independently rated different score of each evaluation index factor, blinded to each other's reactions. The scoring process uses a 9-level scale with scores ranging from 1 to 9 and their reciprocal to represent the relative importance between any two indicators. Constructing judgment matrices for both the criterion and indicator layers allows for the evaluation of spatial demand for EVs PCS. The comparison rule is the relative influence of the two evaluation indicators on the space demand of PCS.

After establishing the judgment matrices for each level of indicators and completing the scoring process, the weights of each indicator level were determined. To guarantee the reasonableness, consistency, and credibility of the scoring results, and to determine their conformity with logical judgment calibration, a consistency test was performed. This paper presents the scores from one expert as a representative example; detailed calculations are demonstrated in Table 5, Table 6, Table 7, Table 8, Tables 9 and 10.

After calculation, the maximum eigenvalue of the consistency test of the judgment matrix of Ui to U indicators is determined as 5.273, and the consistency index C.I. = 0.068. When $n = 5$, R.I. = 1.12. Therefore, C.R. = $0.061 < 0.100$, and it passes the consistency test.

According to the principle of complementarity of the judgment matrix, when there are only two items for the indicators of the second-order judgment matrix, the consistency problem does not exist. Therefore, test in this step is not necessary.

After calculation, the maximum eigenvalue of the consistency test of

Table 5

Weight calculation of the judgment matrix of Ui to U.

U	U1	U2	U3	U4	U5	Weight (Wi)
U1	1	2	1/2	5	5	0.287
U2	1/2	1	1/2	6	3	0.210
U3	2	2	1	6	4	0.379
U4	1/5	1/6	1/6	1	2	0.064
U5	1/5	1/3	1/4	1/2	1	0.060

Table 6

Weight calculation of the judgment matrix of Uij to U1.

U1	U11	U12	Weight (Wi)
U11	1	3	0.750
U12	1/3	1	0.250

Table 7

Weight calculation of the judgment matrix of Uij to U2.

U2	U21	U22	U23	Weight (Wi)
U21	1	3	4	0.615
U22	1/3	1	3	0.268
U23	1/4	1/3	1	0.117

Table 8

Weight calculation of the judgment matrix of Uij to U3.

U3	U31	U32	U33	U34	U35	Weight (Wi)
U31	1	3	5	2	3	0.388
U32	1/3	1	3	1/4	2	0.139
U33	1/5	1/3	1	1/3	1/2	0.065
U34	1/2	4	3	1	4	0.312
U35	1/3	1/2	2	1/4	1	0.096

Table 9

Weight calculation of the judgment matrix of Uij to U4.

U4	U41	U42	Weight (Wi)
U41	1	1/2	0.333
U42	2	1	0.667

Table 10

Weight calculation of the judgment matrix of Uij to U4.

U5	U51	U52	Weight (Wi)
U51	1	1/4	0.800
U52	4	1	0.200

the judgment matrix of Uij to the U2 indicator is determined as 3.074, and the consistency index C.I. = 0.037. When $n = 3$, R.I. = 0.58. Therefore, C.R. = $0.063 < 0.100$, and it passes the consistency test.

After calculation, the maximum eigenvalue of the consistency test of the judgment matrix of Uij to the U3 indicator is determined as 5.256, and the consistency index C.I. = 0.064. When $n = 5$, R.I. = 1.12. Therefore, C.R. = $0.057 < 0.100$, and it passes the consistency test.

According to the principle of complementarity of the judgment matrix, when there are only two items for the indicators of the second-order judgment matrix, the consistency problem does not exist. Thus, test in this step is not necessary.

According to the principle of complementarity of the judgment matrix, when there are only two items for the indicators of the second-order judgment matrix, the consistency problem does not exist. Thus, test in this step is not necessary.

3.3.2. Calculation of comprehensive weights

The composite weight reflects each indicator's contribution to the overall target layer. Having calculated and verified the weights assigned by the 15 experts assessing spatial demand—all of which passed the consistency test—the average weight was adopted for the evaluation of PCS spatial demand for EVs, as presented in Table 11.

The weights assigned to the five dimensions influencing PCS spatial demand are ranked as follows: Infrastructure ($U3 = 0.404$), traffic organization ($U2 = 0.300$), population distribution ($U1 = 0.127$), regional economy ($U5 = 0.115$), and land use ($U4 = 0.054$). This indicates that

Table 11
Weights of evaluation indicators for the spatial demand of PCS based on AHP.

(U)Target level	(Ui)Criterion level	Weight	(Uij)Index level	Weight
(U)Evaluation of spatial demand for PCS of EVs	(U1) Population distribution	0.127	(U11)Population vitality intensity on weekday	0.047
			(U12)Population vitality intensity on rest day	0.080
	(U2) Traffic organization	0.300	(U21)The closeness of road network	0.117
			(U22)The betweenness of road network	0.136
			(U23)Density of road network	0.047
			(U31)The scale of commercial center	0.161
			(U32)The scale of public service facility	0.058
	(U3) Infrastructure	0.404	(U33)The scale of park view	0.043
			(U34)The scale of transportation station	0.120
			(U35)The scale of residential neighborhood	0.022
			(U41)Building coverage	0.031
			(U42)Floor area ratio	0.023
	(U4) Land use	0.054	(U51)Nighttime light distribution	0.091
			(U52)House price level	0.024
	(U5) Regional economy	0.115		

experts and EVs users prefer PCS placement in areas with concentrated infrastructure. Such areas are usually destinations or passageways and are prone to generating charging demands. It is the area with convenient transportation, where users may need to recharge on the way. Population-active areas and economically prosperous areas are also considerations. It indicates that areas with dense population and higher economic levels have greater charging demands. However, the effect of land use is relatively small.

The ranking of the comprehensive weights of the 14 elements of the indicator layer can be obtained in the following order: The scale of commercial center ($U31 = 0.161$) > The betweenness of road network ($U22 = 0.136$) > The scale of transportation station ($U34 = 0.120$) > The closeness of road network ($U21 = 0.117$) > Nighttime light distribution ($U51 = 0.091$) > Population vitality intensity on rest days ($U12 = 0.080$) > The scale of public service facility ($U32 = 0.058$) > Density of road network ($U23 = 0.047$) = Population vitality intensity on weekdays ($U11 = 0.047$) > The scale of park view ($U33 = 0.043$) > Building coverage ($U41 = 0.031$) > House price level ($U52 = 0.024$) > Floor area ratio ($U42 = 0.023$) > The scale of residential neighborhood ($U35 = 0.022$).

3.4. Determination of the weights of evaluation indicators by entropy method

To ensure both objective and accurate weight assignment, this study incorporates an entropy method to evaluate the spatial demand for EVs PCS, drawing upon comprehensive consideration of expert opinions and EV user preferences. To facilitate comparison among indicators with diverse characteristics and dimensions, a Min-Max normalization method was selected to eliminate the effect of dimensions. For the 14 evaluation indicators in each street space unit, entropy values and weights for each indicator were sequentially calculated utilizing Stata

statistical analysis software, as presented in Table 12.

Analysis of Table 11 indicates that the criterion layer weights, derived through the entropy method for evaluating the spatial demand of PCS, follow this descending order: Infrastructure ($U3 = 0.393$), traffic organization ($U2 = 0.204$), land use ($U4 = 0.179$), regional economy ($U5 = 0.174$), and population distribution ($U1 = 0.050$). While these entropy-derived weights are somewhat different from the results obtained through the AHP method, both methods consistently prioritize infrastructure and traffic organization. This prioritization indicates a strong relationship between the spatial layout of EVs PCS and both the distribution of existing infrastructure and the specific construction characteristics of the transportation network. In addition, it highlights the close connection between spatial demand for PCS and the transportation habits, daily routines, and working environments of residents.

The 14 indicator layer factors, ranked by weight, yielded the following hierarchy: The scale of park view ($U33 = 0.167$) > The betweenness of road network ($U22 = 0.149$) > Floor area ratio ($U42 = 0.119$) > House price level ($U52 = 0.102$) > Nighttime light distribution ($U51 = 0.072$) > The scale of commercial center ($U31 = 0.062$) > Building coverage ($U41 = 0.060$) > The scale of public service facility ($U32 = 0.056$) = The scale of residential neighborhood ($U35 = 0.056$) > The scale of transportation station ($U34 = 0.052$) > Density of road network ($U23 = 0.029$) > The closeness of road network ($U21 = 0.026$) = Population vitality intensity on rest days ($U12 = 0.026$) > Population vitality intensity on weekday ($U11 = 0.024$). In comparison to subjective weighting methods, the entropy method indicates significant differences in the individual indicator weights, most notably for the scale of park view, the betweenness of road network, floor area ratio, and house price level. The uncertainty and variability in these factors significantly affect the evaluation results, contributing significantly to their assigned weight proportions and have a significant effect on the evaluation results.

3.5. Calculation of AHP-entropy method weight

The AHP-Entropy method weights for evaluating the spatial demand of PCS for EVs, derived from integrating the index weights calculated separately by AHP and the entropy method, are presented in Table 13.

Applying the AHP-Entropy method, the weights of the criterion layer indicators for evaluating PCS spatial demand are ranked in descending order as follows: Infrastructure ($U3 = 0.403$), traffic organization ($U2 = 0.356$), regional economy ($U5 = 0.129$), land use ($U4 = 0.066$), and population distribution ($U1 = 0.046$). The presence of infrastructure at destinations and the transportation network facilitating movement are the two primary factors influencing PCS spatial demand. In addition, the regional economy's developmental state plays a crucial role. The relatively high weights assigned to land use indicators are attributed to their substantial data dispersion and rich informational value; whereas, the regional vitality intensity, reflecting the daily distribution of the population, has become the factor with the smallest proportion.

The weights of the 14 factors at the indicator layer indicate the following hierarchical structure: The betweenness of road network ($U22 = 0.292$) > The scale of commercial center ($U31 = 0.144$) > The scale of park view ($U33 = 0.104$) > Nighttime light distribution ($U51 = 0.094$) > The scale of transportation station ($U34 = 0.090$) > The scale of public service facility ($U32 = 0.047$) > The closeness of road network ($U21 = 0.044$) > Floor area ratio ($U42 = 0.039$) > House price level ($U52 = 0.035$) > Population vitality intensity on rest days ($U12 = 0.030$) > Building coverage ($U41 = 0.027$) > Density of road network ($U23 = 0.020$) > The scale of residential neighborhood ($U35 = 0.018$) > Population vitality intensity on weekday ($U11 = 0.016$).

4. Discussion

This paper constructs a five-dimensional evaluation index system for assessing the spatial demand for EVs PCS, using the AHP-Entropy

Table 12

Weights of evaluation indicators for the spatial demand of PCS based on entropy method.

(U)Target level	(Ui)Criterion level	Weight	(Uij)Index level	Entropy	Weight
(U)Evaluation of spatial demand for PCS of EVs	(U1) Population distribution	0.050	(U11)Population vitality intensity on weekday	0.975	0.024
			(U12)Population vitality intensity on rest day	0.973	0.026
	(U2) Traffic organization	0.204	(U21)The closeness of road network	0.972	0.026
			(U22)The betweenness of road network	0.846	0.149
			(U23)Density of road network	0.970	0.029
	(U3) Infrastructure	0.393	(U31)The scale of commercial center	0.936	0.062
			(U32)The scale of public service facility	0.942	0.056
			(U33)The scale of park view	0.827	0.167
			(U34)The scale of transportation station	0.947	0.052
			(U35)The scale of residential neighborhood	0.942	0.056
	(U4) Land use	0.179	(U41)Building coverage	0.938	0.060
			(U42)Floor area ratio	0.879	0.119
	(U5) Regional economy	0.174	(U51)Nighttime light distribution	0.925	0.072
			(U52)House price level	0.894	0.102

Table 13

AHP-Entropy method weights for spatial demand evaluation of PCS for EVs.

Evaluation dimension	Weight	Evaluation indicators	Correlation	AHP weight	Entropy method weight	AHP-Entropy method weights
(U1) Population distribution	0.046	(U11)Population vitality intensity on weekday	+	0.047	0.024	0.016
		(U12)Population vitality intensity on rest day	+	0.08	0.026	0.030
(U2) Traffic organization	0.356	(U21)The closeness of road network	+	0.117	0.026	0.044
		(U22)The betweenness of road network	+	0.136	0.149	0.292
		(U23)Density of road network	+	0.047	0.029	0.020
(U3) Infrastructure	0.403	(U31)The scale of commercial center	+	0.161	0.062	0.144
		(U32)The scale of public service facility	+	0.058	0.056	0.047
		(U33)The scale of park view	+	0.043	0.167	0.104
		(U34)The scale of transportation station	+	0.12	0.052	0.090
		(U35)The scale of residential neighborhood	+	0.022	0.056	0.018
(U4) Land use	0.066	(U41)Building coverage	+	0.031	0.060	0.027
		(U42)Floor area ratio	+	0.023	0.119	0.039
(U5) Regional economy	0.129	(U51)Nighttime light distribution	+	0.091	0.072	0.094
		(U52)House price level	+	0.024	0.102	0.035

method to determine index weightings. The analysis shows that all factors significantly influence the evaluation. Based on these findings and comparisons with existing literature, the paper proposes targeted five-dimensional spatial layout strategies for EVs PCS, which are discussed in detail.

4.1. Analysis of the results and comparison with existing studies

Discussion 1: Influential Factors in the Evaluation of Spatial Demand for EVs PCS. Road network betweenness carries the greatest weight among the influential factors. Most scholars have emphasized the crucial role of road network accessibility in promoting EVs adoption and usage. Relevant research suggests that strong road network connectivity, specifically good crossing performance, directly improves both the accessibility of PCS and the convenience afforded to users, thereby increasing the frequency of PCS utilization. Moreover, existing literature indicates that areas of commercial concentration and park areas tend to attract high volumes of pedestrian and vehicular traffic (He et al., 2020b). The setting of PCS can effectively meet the charging needs of users in high-density areas. This observation is consistent with the findings and highlights the necessity of prioritizing commercial and leisure areas in PCS layout planning (Wei et al., 2023). In addition, the relevant literature shows that the nighttime light distribution is not only related to urban activity, but also closely related to the charging demand of users in different time periods. Transportation hub is an important node for people to travel. The scale of public service facilities is closely related to the daily life of residents, and a good living environment can significantly improve the charging experience of EVs users.

Discussion 2: The effect of spatial demand evaluation dimension of EVs PCS. The infrastructure is able to affect the layout and use of PCS to

the greatest extent and is consistent with existing literature. This finding highlights the central role of adequate and efficient infrastructure in supporting the adoption of EVs. Second, with respect to the effect of traffic organization, relevant literature points out that reasonable traffic flow and organization can reduce charging waiting time and improve charging efficiency; where the relationship between traffic mobility and the layout of charging facilities are highlighted in this study. It demonstrates that reasonable traffic organization strategies can significantly enhance users' charging experience. In addition, regional economy also represents a significant aspect in this study. Relevant literature mentions that areas with a high level of economic development usually have a stronger demand for EVs and PCS and are confirmed by this study, and it is discovered that the concentration of regional economic activities is closely related to the demand for charging. Moreover, the effects of land use and population distribution cannot be ignored and it should be noted that existing literature demonstrates that the setting of PCS should consider the nature of land use and population density (Li et al., 2015). Meanwhile, it is also necessary emphasized that reasonable land use planning is capable of effective promotion of the layout of PCS, hence it can better serve users. In general, the five dimensions of infrastructure, traffic organization, regional economy, land use and population distribution exert a joint affect on the layout and use of EVs PCS. Accordingly, this study not only aligns with the existing literature, but also offers a more comprehensive perspective and practical basis for future PCS planning.

Discussion 3: The three methods show some differences in the analysis of the spatial demand weight of EVs PCS, as showed in Fig. 6. As a subjective method, AHP mainly evaluates the relative importance of indicators based on expert analysis and user needs. Its results reflect public preferences but may have subjective randomness. The entropy

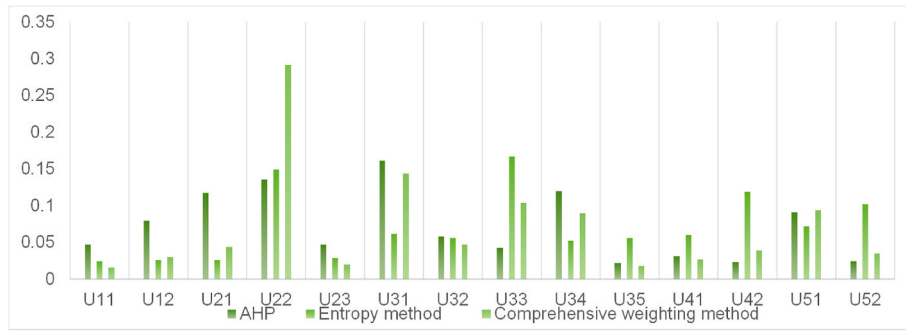


Fig. 6. Comparison of index weight calculation results.

method calculates weights according to the dispersion of objective data, yet it might overlook the subjective importance of indicators to users' use needs. Thus, the results of the two methods differ. The AHP-entropy method not only considers the actual demand of expert users, but also considers the actual data information, avoiding the incompleteness of single-method results. For example, "the scale of park view" has a low weight in AHP as experts and users consider it less significant for PCS demand compared to other factors. However, its entropy value is high because park views are unevenly distributed in urban space, and a single park place has a large space. A single park often attracts tourists and leisure activities and has potential charging needs at specific times or for certain groups.

4.2. Discussion on the spatial layout of PCS for EVs

4.2.1. Population distribution dimension

Strategy 1: In high-density urban areas, the deployment of compact, high-efficiency PCS should be given precedence. Slower PCS should be positioned in surrounding less dense areas, and in the interstitial zones between high-density areas to optimize traffic management and accommodate both immediate and projected charging demands. In central business districts and commercial centers, the emphasis should be on rapid PCS installations, supplemented by slower PCS to serve the needs of office workers. Residential areas should primarily offer slower PCS to facilitate overnight charging. Transportation hubs and major thoroughfares require a significant number of rapid PCS. Public venues such as hospitals and schools should incorporate PCS installations scaled to their population density, while tourist destinations should also offer adequate charging infrastructure to meet visitor requirements as shown in Fig. 7.

Strategy 2: For low-density areas on the urban periphery, PCS deployment should be scaled accordingly, with careful consideration given to both placement and charger type. Strategies for shared PCS use among neighboring areas can be implemented, supplemented by mobile charging units and distributed charging infrastructure. Rapid PCS should be located at key transportation nodes, while slower PCS should be deployed across residential, commercial, and industrial sectors. The integration of intelligent services, including online booking and mobile payment options, can significantly improve user experience as shown in Fig. 8.

4.2.2. Traffic organization dimension

Strategy 1: Install Fast PCS along arterial roadways and at key transportation interchanges offering convenient parking and transfer possibilities. Suitable locations include commercial districts, office complexes, petrol stations, areas beneath overpasses, and major intersections.

Strategy 2: Position Slow PCS along secondary roadways reflected by reduced traffic volume, lower traversability, and high accessibility. Appropriate sites include residential neighborhoods, parks, and public transit stations. This approach will accommodate daily commutes and reduce pressure on main roads.

Strategy 3: Enhancing accessibility involves increasing PCS in high-traffic areas while narrowing service ranges in less utilized spots. It is essential to connect PCS with public parking lots, residential areas, commercial and office areas, industrial parks, tourist attractions, important commuting routes, transportation nodes, large sports stadiums, as well as important areas such as schools and hospitals as shown in Fig. 9.

Strategy 4: Implement staggered PCS operating hours to more

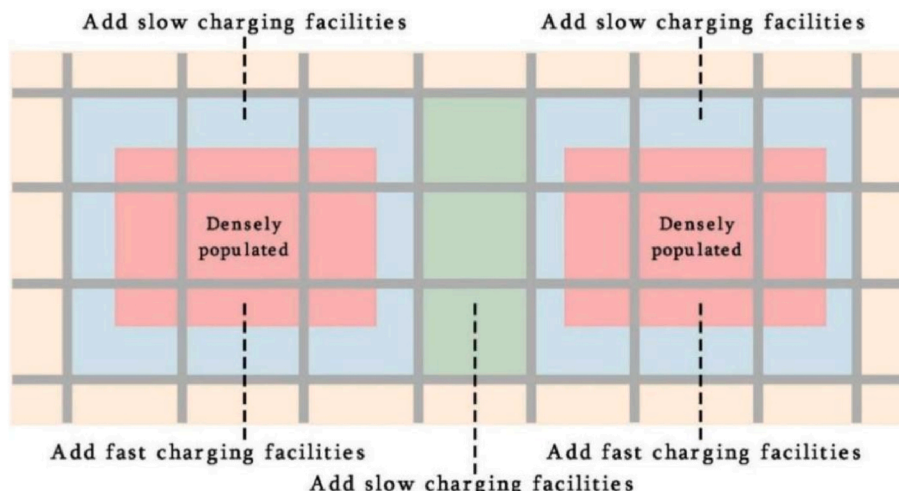


Fig. 7. Layout of PCS in densely populated areas and surrounding areas.

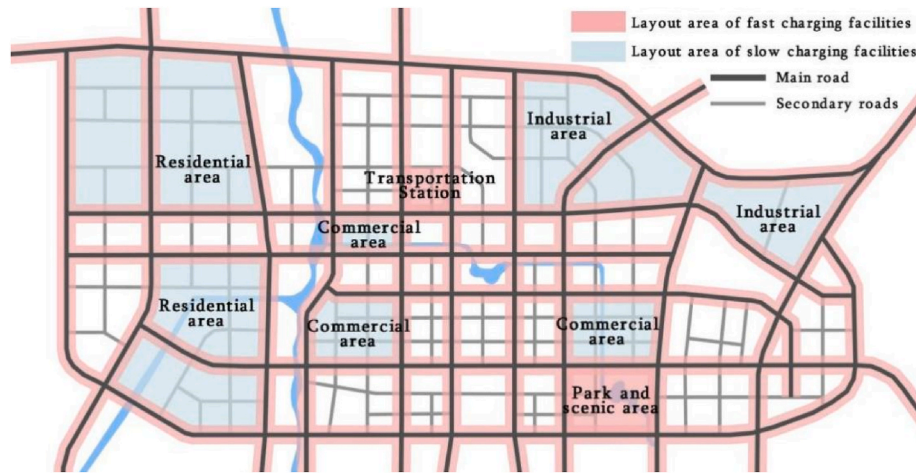


Fig. 8. Layout of PCS in different functional areas in sparsely populated areas.

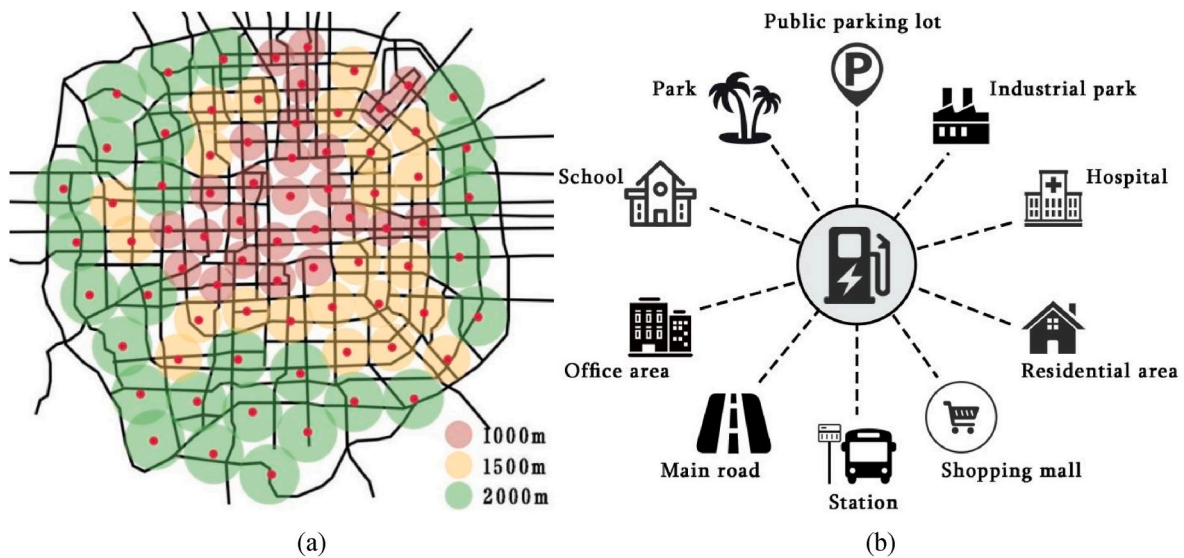


Fig. 9. PCS layout strategy from the perspective of traffic organization.

Note: (a) The service range of PCS in different regions is different; (b) Connection of PCS to important public spaces.

effectively manage peak demand. PCS availability should be limited during peak periods and extended during off-peak periods.

Strategy 5: In low-density areas, adding PCS or improving road networks through new constructions and optimizations can be beneficial, including new road construction, road widening, and traffic signal optimization. Clear signage and integration with navigation systems will help drivers locate nearby PCS efficiently.

4.2.3. Infrastructure dimension

In large shopping malls and commercial buildings, fast PCS should be integrated with underground and above-ground parking. PCS location should focus on entrances and exits. In hospitals, fast PCS should be near emergency room entrances, while slow PCS serve patient and visitor parking. Schools should have fast PCS in parking lots, staff dormitory areas, teaching buildings and main campus entrances and exits, with slow PCS near teaching and dormitory buildings. Emergency PCS can be placed in public venues like stadiums and libraries, with dedicated charging areas in stadium parking lots. Government agencies should install appropriate fast and slow PCS in office building parking lots and public areas. Community cultural centers need slow PCS near parking and entrances. Parks and scenic spots should primarily have slow PCS in large parking areas, with some fast PCS at entrances. During peak tourist

seasons, temporary mobile charging stations should be added, along with real-time PCS information via mobile apps. Transportation hubs like railway stations and expressway service areas should have fast PCS at key locations along main roads. New residential communities should include slow PCS in garages, along roads, and in community centers, while old communities can repurpose green belts and parking areas for slow PCS (Hao, 2019).

4.2.4. Land use dimension

Strategy 1: For underutilized public lands, including vacant lots, green spaces, and abandoned industrial areas, PCS should be introduced through land consolidation and functional transformation. For instance, old industrial sites can be converted into EVs parks that integrate PCS with vehicle sales and maintenance. Similarly, outdated commercial and residential areas can be transformed into multifunctional service hubs for charging, parking, and commerce. In urban-rural fringes and areas undergoing urban renewal, advance planning for PCS construction is essential.

Strategy 2: In downtown, including commercial districts and high-density residential areas, three-dimensional, intelligent integrated parking and PCS can be established, utilizing underground parking lots and existing public facilities like streetlight and electric poles.

4.2.5. Regional economic dimension

Fast PCS should be strategically located along major thoroughfares in industrial clusters and near public transportation hubs. Both fast and slow charging stations should be positioned in public parking lots, underground garages, and on the top floors of commercial buildings. In high-end residential, commercial, and tourism areas, increasing the quantity and density of efficient PCS is essential, offering services like reservation charging and online payments, along with comfortable rest areas. High-end residential communities should focus on slow PCS, while commercial centers can accommodate both fast and slow charging. For tourist areas, PCS should align with tourist routes, located in hotels and service centers. In low-consumption areas, PCS quantity and density can be reduced, focusing on cost-effective solutions.

4.2.6. Optimization strategies adjustment with EVs technology advancements

With the continuous evolution of EVs technology like enhanced driving range, increasing charging speed, the PCS layout strategy requires adjustment. In high-density areas, considering increased mileage, the number of fast PCS can be reduced while slow PCS increased. Low-density areas should cut mobile PCS reliance, optimize fixed ones, and transfer resources to the urban edge with unmet charging demands. Main roads and traffic hubs can enhance fast charge power and adjust spacing. The fast charging ratio should be increased and the slow charging ratio should be reduced to make the facility layout closer to the demand point. In commercial centers, high-power fast PCS can be built to meet the demand for rapid power replenishment. For wireless charging, embedded equipment should be planned, such as laying coils under busy roads in urban areas. With autonomous driving, automated PCS can be built on city edge land, and PCS can be centrally arranged in shared car hubs according to changes in travel patterns. It could improve user charging experience and promote EVs industry sustainability.

5. Conclusion

Considering the increasing importance of environmental protection and sustainable development, EVs, as clean energy transportation solutions, have become a central focus for the future of transportation. The widespread adoption of EVs necessitates located PCS, a process requiring careful evaluation of numerous contributing factors. This study, utilizing Jinan, China as a case study, employs the AHP-Entropy method to appraise the key factors influencing the spatial demand for EVs PCS. It identifies the relative importance and weighting of these factors, thereby offers an analytical framework for optimizing PCS placement. First, the study establishes an evaluation index system including five dimensions: population distribution, traffic organization, infrastructure, land use, and regional economic activity. Then, the AHP-Entropy method is applied to assess and analyze the various contributing factors. The findings indicate the following: 1) The prioritized weighting of criteria layer indicators derived from the combined weighting method is: infrastructure (0.403), traffic organization (0.356), regional economy (0.129), land use (0.066), and population distribution (0.046). 2) Of the 14 factors at the indicator level, the betweenness of road network (0.292) represents the most influential factor, followed by the scale of commercial centers (0.144) and the scale of park views (0.104). Nighttime light distribution (0.094) and the scale of transportation stations (0.090) are also identified as important considerations. By integrating the AHP and entropy methods, this study leverages objective data while incorporating user preferences, resulting in more robust and comprehensive evaluation outcomes. This combined approach effectively addresses the limitations of relying on a single analytical method and offers a more objective and practical optimization strategy for PCS placement.

This paper develops a spatial demand evaluation system for PCS using multidimensional data, providing a reference for future urban research. The findings support improving PCS utilization and service

quality, and help optimize PCS placement to address regional demand differences and reduce uneven distribution. By prioritizing high-demand areas such as transit centers and commercial districts, transportation planners can improve resource efficiency, expand coverage, and boost EV adoption. However, the study is limited to Jinan City, which may affect its generalizability to other urban contexts. Considering differences in economic development, population density, and traffic infrastructure across cities, the transferability of these conclusions requires further analysis in other locations. In addition, limited data on some of the influencing factors may result in some key factors not being fully analyzed. In conclusion, this study offers valuable decision-making support for the spatial layout of PCS. As additional data becomes available, the method presented here can be adapted to other cities, offering broader support for EVs PCS development and presenting a novel approach to achieving low-carbon emission reductions.

CRediT authorship contribution statement

Qimeng Ren: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ming Sun:** Writing – review & editing, Visualization, Supervision, Resources, Project administration, Methodology, Funding acquisition, Data curation, Conceptualization.

Informed consent statement

Informed consent was obtained from all subjects involved in the study.

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Declaration of competing interest

The reviewer declared a shared affiliation with the authors to the handling editor at the time of review.

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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